

DATA ARTICLE OPEN ACCESS

LandFrag: A Dataset to Investigate the Effects of Forest Loss and Fragmentation on Biodiversity

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ABSTRACT

Motivation: The accelerated and widespread conversion of once continuous ecosystems into fragmented landscapes has driven ecological research to understand the response of biodiversity to local (fragment size) and landscape (forest cover and fragmentation) changes. This information has important theoretical and applied implications, but is still far from complete. We compiled the most comprehensive and updated database to investigate how these local and landscape changes determine species composition, abundance and trait diversity of multiple taxonomic groups in forest fragments across the globe.

Main Types of Variables Contained: We gathered data for 1472 forest fragments, providing information on the abundance and composition of 9154 species belonging to vertebrates, invertebrates, and plants. For 2703 of these species, we obtained more

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than 20 functional traits. We provided the spatial location and size of each fragment and metrics of landscape composition and configuration.

Spatial Location and Grain: The dataset includes 1472 forest fragments sampled in 121 studies from all continents except Antarctica. Most datasets (77%) are from tropical regions, 17% are from temperate regions, and 6% are from subtropical regions. Species abundance and composition were collected at the plot or fragment scale, whereas the landscape metrics were extracted with buffer size ranging from a radius of 200–2000 m.

Time Period and Grain: Data on the abundance of species and community composition were collected between 1994 and 2022, and the landscape metrics were extracted from the same year that a given study collected the abundance and composition data.

Major Taxa and Level of Measurement: The studied organisms included invertebrates (Arachnida, Insecta and Gastropoda; 41% of the datasets), vertebrates (Amphibia, Squamata, Aves and Mammalia; 44%), and vascular plants (19%), and the lowest level of identification was species or morphospecies.

Software Format: The dataset and code can be downloaded on Zenodo or GitHub.

1 | Introduction

The growing expansion of agriculture and infrastructure is causing the annual loss of millions of hectares of forest worldwide, especially in the tropics (Global Forest Watch 2024). These massive land-use changes have caught the attention of ecologists and conservation biologists because they are shaping ecological patterns and processes across different spatial scales (Lôbo et al. 2011; Tscharrntke et al. 2012; Haddad et al. 2015; Arroyo-Rodríguez et al. 2020; Hansen et al. 2020). In fact, ecological theory postulates that in human-modified landscapes, the structure of species assemblages can depend on both local—(e.g., fragment size) and landscape-scale factors (e.g., forest cover and fragmentation), but their relative role remains debated (e.g., Fletcher et al. 2018; Fahrig et al. 2019). For instance, the habitat amount hypothesis predicts that species density in a given site is more strongly and positively related to the amount of available habitat in the surrounding landscape than to the size of the habitat fragment within which the site is located. Although some studies support this prediction (Watling et al. 2020), others suggest that species density is weakly related to habitat amount (e.g., Martínez-Ruiz et al. 2024). Similarly, the response of species to fragmentation seems to be generally weak, but positive and negative responses have also been frequently reported (Fahrig 2017). Therefore, further research is required to ascertain the primary drivers of biodiversity loss in human-modified landscapes, the species and/or group of species that are most susceptible to forest spatial changes, and the landscape and/or regional contexts under which biodiversity responses become predominantly negative (Fletcher et al. 2018; Fahrig et al. 2019).

Global reviews and meta-analyses can be valuable approaches to obtain a more comprehensive understanding of the response of biodiversity to spatial changes of the habitat. To this end, an increasing number of global datasets have been compiled in the last decade (e.g., BIOFRAG: Pfeifer et al. 2014; PREDICT: Hudson et al. 2014; FragSAD: Chase et al. 2019). These datasets have been useful to test important ecological hypotheses, such as the ‘ecosystem decay hypothesis’ (Chase et al. 2020), and the ‘extinction filter hypothesis’ (Betts et al. 2019; Weeks et al. 2023). BIOFRAG primarily contains data on the presence/absence of populations (single species) or communities (multiple species) in forest fragments. Although it contains the spatial locations of the fragments, it does not provide species abundances to calculate diversity metrics within fragments, estimates of fragment

size for all studies or information about the structure of the surrounding landscape (Pfeifer et al. 2014). Also, BIOFRAG is not currently open access. In turn, FragSAD is open access and provides information on species abundances and fragment size for all fragments (Chase et al. 2019, 2020), which allows an explicit evaluation of the effects of sampling effort on diversity (Chase et al. 2020). However, it does not include the exact spatial location of the fragments or plots, which prevents the assessment or control of the effects of spatial gradients, as well as the more precise description of the surrounding matrix. To date, most landscape studies using multiple taxonomic groups have relied on complex metrics of landscape structure, such as the ‘edge influence’ index (Betts et al. 2019; Weeks et al. 2023), indirect measurements of matrix quality (e.g., matrix age and matrix type; Chase et al. 2020), or a few configuration metrics (e.g., mean fragment size; Riva and Fahrig 2023).

Here, we gathered a global database of the composition and abundance of 9154 species sampled in 1472 forest fragments as part of 121 studies of different taxonomic groups in tropical, subtropical, and temperate regions (Figure 1). For 2703 of these species, we were also able to compile information on morphological, trophic, habitat, and reproductive traits. We recorded the spatial location of each study fragment, its size (in hectares), and the spatial structure of the landscape surrounding each fragment. In particular, we calculated 10 landscape variables (Table 1) that have been at the core of important ecological debates (e.g., Fahrig et al. 2022). Therefore, this dataset has broad applicability in ecological research, and can potentially be used to address many research questions in landscapes with fragmented forests at all spatial scales, from local to global. Examples of potential research include, but are not limited to, testing the relative importance of fragment- and landscape-related variables affecting biodiversity changes, and determining the traits of species which are the ‘winners’ and ‘losers’ following forest loss and fragmentation (see, e.g., Riva and Fahrig 2023; Pinho et al. 2024; Zhang et al. 2024). Importantly, since the effect of landscape structure on biodiversity can go undetected if the structure is assessed at the wrong scale (Jackson and Fahrig 2015), we calculated landscape metrics in circular landscapes of different sizes (from 200 to 2000 m radius). This multiscale information can be highly valuable to identify the so-called ‘scale of effect’ of each landscape metric on each response (Jackson and Fahrig 2015)—an emerging topic in landscape ecology that can

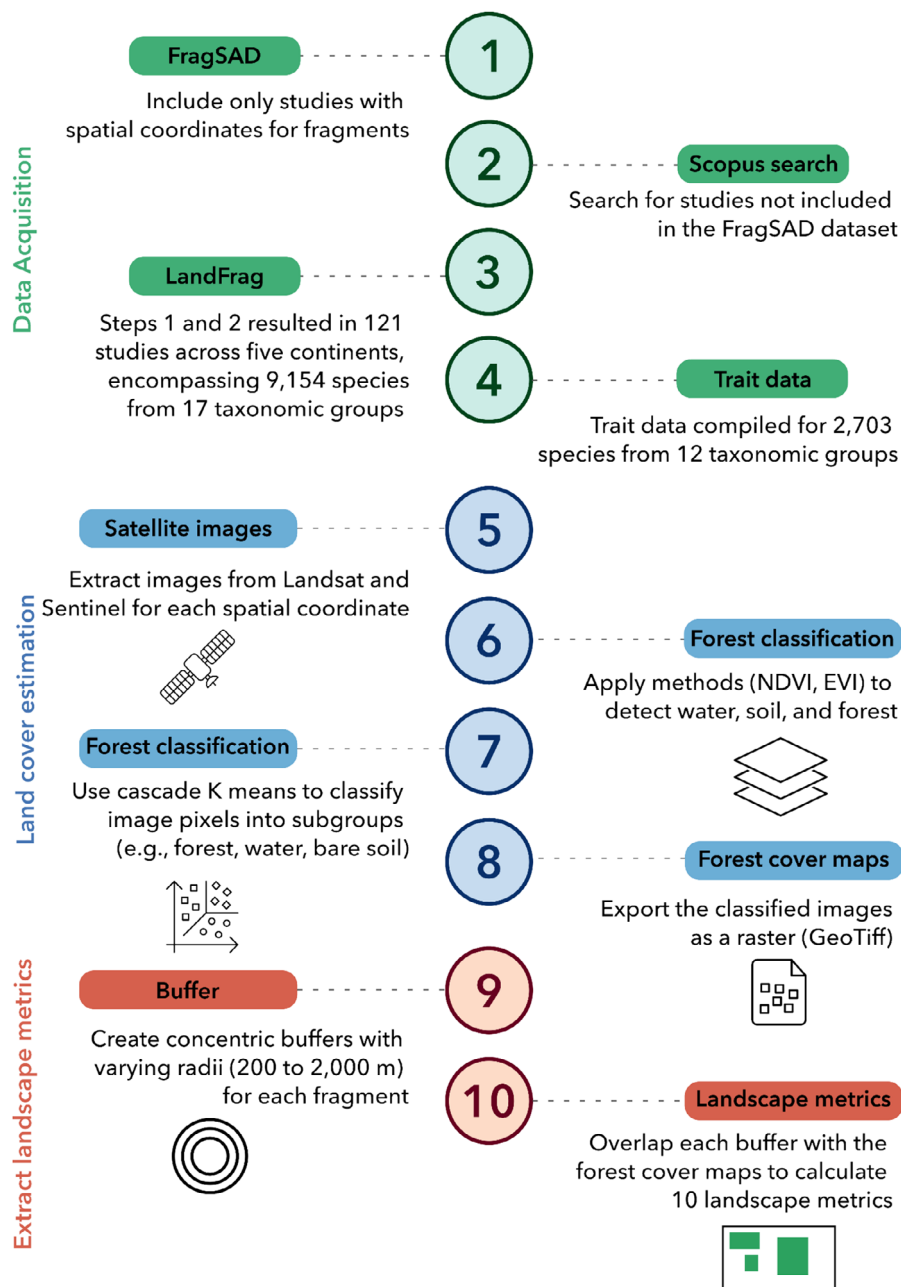


FIGURE 1 | Conceptual figure summarising the 10 most important steps to collect species and trait data (data acquisition, green), obtain satellite image and estimate forest cover (land cover estimation, blue), and to extract the landscape metrics (red).

be used for assessing important hypotheses on spatial scaling issues (Miguet et al. 2016).

2 | Methods

2.1 | Data Acquisition

We used FragSAD (Chase et al. 2019) as the starting point of this new dataset. While FragSAD includes species abundance and composition in forest fragments, it does not provide the exact spatial location of each surveyed forest fragment. Thus, we first revisited all studies in the FragSAD and selected only those that collected species in at least four forest fragments; we also excluded those from non-forest habitats such as grasslands. We retrieved

the spatial location (latitude and longitude) of all fragments in a study by: (i) extracting the information from the article (tables) or maps with the coordinate system or (ii) contacting the first or corresponding author(s) of the original papers to request the spatial locations of the fragments. When we extracted data from maps, we used Google Earth to obtain the coordinates from the centroid of the studied fragment. In addition, we included data published since the original 2019 FragSAD publication by performing a new search on Scopus using the same keywords from Chase et al. (2019): ('habitat fragment*' OR 'habitat loss' OR 'forest fragment*' OR 'forest loss' OR 'fragment area' OR 'fragment size' OR 'island area') AND ('diversity' OR 'species diversity' OR 'species richness' OR 'abundance') from 2019 to 2022. For this search, we included those studies with open-access data, and contacted authors listed in the search to ask for published or unpublished datasets

TABLE 1 | Forest landscape metrics describing composition and configuration categories, including area, edge effects, aggregation and shape complexity.

Metric category	Metric group	Landscape metric	Description	Units
Composition	Area and edge	Mean area of fragments	Mean of all fragment areas of forest in the buffer	Hectares
Composition	Area and edge	Percentage of forest in the landscape	Total area of forest in the buffer, divided by the total area of the buffer multiplied by 100	Percent
Configuration	Aggregation	Number of fragments	Number of forest fragments and continuous forests in the landscape	Number
Configuration	Aggregation	Fragment density	Number of fragments in the buffer, divided by the total area of the buffer multiplied by 10,000 and 100	Number/100 ha
Configuration	Aggregation	Inter-fragment isolation distance	Distance from each fragment to its nearest adjacent fragment. The mean distance is then calculated using the distance for all fragments pairs in the landscape	Meters
Configuration	Aggregation	Aggregation index	It equals the number of like adjacencies divided by the theoretical maximum possible number of like adjacencies for that class	None
Configuration	Area and edge	Edge density	Total length of edge (m) in the buffer, divided by the total area of the buffer (m ²) multiplied by 10,000	Meters per ha
Configuration	Area and edge	Largest patch index	Area (m ²) of the largest fragment in the buffer, divided by the total area of the buffer multiplied by 100	Percent
Configuration	Shape	Perimeter-area ratio	Perimeters (m) of the fragment divided by the area (ha) of the fragment	Meters per ha
Configuration	Shape	Contiguity index	Contiguity value of a pixel in a given fragment divided by the area of that fragment	None

fitting our criteria. We called this dataset ‘LandFrag’, which stands for ‘LANDscape and FRAGment data’. The LandFrag dataset gathered information of 1472 forest fragments derived from 121 studies. These studies gathered data from 679,928 individuals, representing 9154 species, documented across five continents and 32 countries (Figures 1 and 2; Supporting Information 1 and 2). These studies were primarily conducted in tropical regions, with 93 out of 121 studies focusing on these areas, and 56% of the studies being in South America. The prominence of fragmentation research in tropical areas has been demonstrated in other datasets (e.g., Pfeifer et al. 2014; Chase et al. 2019). Therefore, conclusions about global patterns must be approached with caution.

We combined a list of all species from these studies into the following taxonomic groups:

- Invertebrates (51 datasets):
 - Arthropoda: Arachnida—Araneae and Opiliones, Coleoptera, Diptera, Hemiptera, Hymenoptera—Formicidae, bees and wasps, Isoptera, Lepidoptera, Orthoptera; and Mollusca: Gastropoda.
- Vertebrates (55 datasets):
 - Amphibia, Aves, Mammalia, Squamata
- Vascular plants (19 datasets)

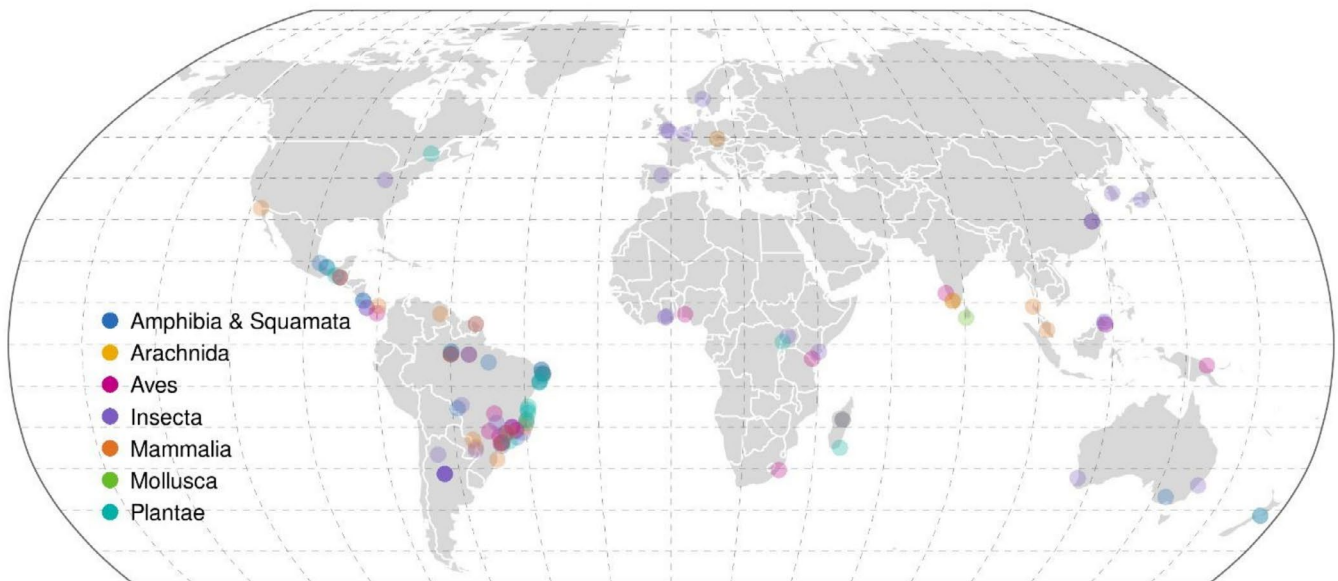


FIGURE 2 | Global distribution of the 121 habitat loss and fragmentation studies organised by taxonomic groups.

2.2 | Aggregating Studies by Sampling Design

Because the sampling design can affect our ability to compare species diversity across fragments and studies (Chase et al. 2020), we classified the 121 studies into three major categories depending on their sampling methods: (i) studies with standardised samples, where the same sampling effort was established in different fragments regardless of their size; (ii) studies using different efforts, usually proportional to fragment area, but which could be standardised by sample size (i.e., number of individuals) because they provide information on the number of individuals per fragment; and (iii) studies with a pooled design, where the total number of individuals of each species is reported for each fragment, but there is no information on the effort (e.g., number of samples) per fragment (see also Chase et al. 2020).

2.3 | Trait Data

We compiled functional traits for 2703 species (29% of all collected species; Table 2; Supporting Information 1). We were not able to provide trait data for 71% of the species because they were (1) unstudied organisms in our collected database, and (2) unidentified species (e.g., morphospecies identified at the class or family level). We used published datasets (Wilman et al. 2014; Oliveira et al. 2017; Kattge et al. 2020; Pinho et al. 2021; Shirey et al. 2022; Tobias et al. 2022) or the original paper describing a given species or genus to obtain trait data. Specifically, for those organisms with organised databases (Gastropoda, Hymenoptera, Lepidoptera, Amphibia, Aves, Mammalia and Plants), we matched the list of species found in the LandFrag dataset with species included in trait databases. For all other taxonomic groups, we searched for trait information in the manuscript (or monograph) that originally described each species (Table 2). For animals, most of the selected traits are associated with morphological features, trophic levels, habitat type, foraging time, and reproductive mode. For plants, the most common traits were dispersal syndrome, leaf Nitrogen, wood density, seed mass and leaf dry mass (Table 2).

2.4 | Estimating Land Cover

We used Google Earth Engine to classify images and estimate land cover for each plot or fragment. We applied cloud mask functions to generate a median image based on the sampling year (or study publication year when the sampling date was not available) for each plot/fragment. The collections used to extract these images were Landsat 4, 5, 7, 8 and Sentinel 2. The datasets consist of atmospherically corrected surface reflectance data produced by the sensors TM (Landsat 4 and 5), ETM+ (Landsat 7), OLI/TIRS (Landsat 8), and the Sentinel MSI sensor. For automatization, we excluded Landsat 7 when possible to ensure extracting information with no interference caused by specific limitations on this satellite (Wijedasa et al. 2012). We used different algorithms depending on the year in which each sampling was carried out. For studies performed before 2013, we selected image collections from Landsat 4 and 5. When there was no image available for a given location, we selected images from Landsat 7. We selected Landsat 8 for studies performed between 2013 and 2015, and Sentinel 2 for studies performed after 2015. Because the studies were conducted in different years, we were unable to use images from the same satellite.

After extracting the images for all plots or fragments, we applied normalised bands of MNDWI (Modified Normalised Difference of Water Index), NDVI (Normalised Difference Vegetation Index) and EVI (Enhanced Vegetation Index) (Wang et al. 2019) to detect water and live green vegetation that will further be used to estimate forest cover. We also used satellite bands of Short Wave Infrared (SWIR) to detect bare soil, near infra-red (NIR) for vegetation and Blue bands for water and shadows. In addition, a mask cloud function was also implemented to prevent interference caused by cloud and shadow in the final forest classification (Anzalone et al. 2024). We selected the classifier Cascade K-means that is based on an unsupervised method of cluster analysis where it splits the image pixels according to their datasets given random trained samples and separates them into subgroups (Caliński and Harabasz 1974). The robustness

TABLE 2 | Trait information and availability for all taxonomic groups in the LandFrag database.

Major taxa	Minor taxa	Trait information	Trait names	Number of species with trait information	Source
Arachnida	Araneae	Not available	None	None	None
Arachnida	Opiliones	Available	Body length and width, femur IV length, habitat	13	Manuscripts containing the original species description
Coleoptera	—	Not available	None	None	None
Diptera	—	Available	Body size, trophic level, reproductive mode, foraging strat, larvae diet, larvae trophic level	46	Manuscripts containing the original species description
Gastropoda	—	Available	Age at maturity, longevity, clutch size, maximal shell size, survival of dry period, inundation tolerance	27	Ellers et al. 2018
Hemiptera	—	Not available	None	None	None
Hymenoptera	Aculeata	Not available	None	None	None
Hymenoptera	Formicidae	Available	Morphological traits (e.g., head width and length, clypeus length, mandible length) and dominant colour	26	Parr et al. 2017
Hymenoptera	Parasitic wasps	Available	Body size, forewing length, diet, reproductive mode and foraging strata	100	Manuscripts containing the original species description
Isoptera	—	Not available	None	None	None
Lepidoptera	—	Available	Wingspan, forewing length, voltinism.	279	Shirey et al. 2022
Orthoptera	—	Available	Pronotum and femur size	75	Manuscripts containing the original species description
Amphibia	—	Available	Body size and mass, foraging habitat and period and diet	227	Oliveira et al. 2017 and manuscripts containing the original species description
Aves	—	Available	Morphological variables (e.g., beak length, body mass, tarsus length), habitat, trophic level	234	Tobias et al. 2022
Mammalia	—	Available	Body mass, diet, foraging strata and activity period, lifestyle	262	Wilman et al. 2014
Squamata	—	Available	Maximum snout to vent length, diet, substrate, reproductive mode, activity period	94	Manuscripts containing the original species description
Plants	—	Available	Dispersal syndrome, leaf nitrogen, wood density, seed dry mass, leaf dry mass	1320	(Kattge et al. 2020; Pinho et al. 2021)

and efficiency of this method in handling large datasets are better when compared to the traditional *K*-means classifier (Deka and Saha 2023). To train the models, we generated 400,000 random pixels in the landscape of interest using all normalised bands. We used this number of random pixels because it is a nice balance of the trade-offs among sample size, computational cost, and classification accuracy. We exported all files (i.e., forest cover maps) as GeoTiff, and we used them to calculate the landscape metrics.

2.5 | Extracting Landscape Variables at Different Scales

One way to investigate the effect of forest loss and fragmentation (Figure 3A) is to extract the landscape metrics surrounding the fragments. To do that, we established concentric buffers (Figure 3B) of 200–2000 m, in 100 m increments. These buffers were created from the centre of each sampled plot/fragment for each study site using the GeoTiff files obtained from the image classification algorithm. This multiscale approach is relevant to select the most appropriate scale affecting different species or taxonomic groups. We fit the previously created binary raster to each buffer and projected these rasters to the WGS84 UTM zone corresponding to each plot/fragment. We used these buffers and the forest cover maps to extract the following landscape metrics: forest cover, fragment density, number of fragments, forest edge

density, largest patch index, contiguity index, euclidean nearest-neighbour distance, proximity index and perimeter. We describe each metric along with its category, group, scale, units and an ecological meaning in Table 1. We provided a non-exhaustive list of landscape metrics, but users can extract other metrics available in the *landscapemetrics* package (Hesselbarth et al. 2019) by adapting the code provided (Supporting Information).

Studies spanned the full gradient from intact (e.g., a buffer hosting a single forest fragment within a landscape with 100% forest cover) to intensively fragmented landscapes (e.g., landscapes with low forest cover and a large number of very small forest fragments). This whole gradient was likewise found for all taxonomic groups, allowing for comparison both within and across them (Figure S1). Users can also use landscape metrics at different scales to investigate how the scale of effect can change within and between taxonomic groups or ecosystems (Jackson and Fahrig 2015). We summarised all steps to create the LandFrag dataset in Figure 1; Supporting Information 2.

3 | Value and Potential Uses of the LANDFRAG Dataset

The functioning and ecological integrity of natural ecosystems largely depend on biodiversity and the ecological traits of the species that inhabit them (Andresen et al. 2018). Therefore,

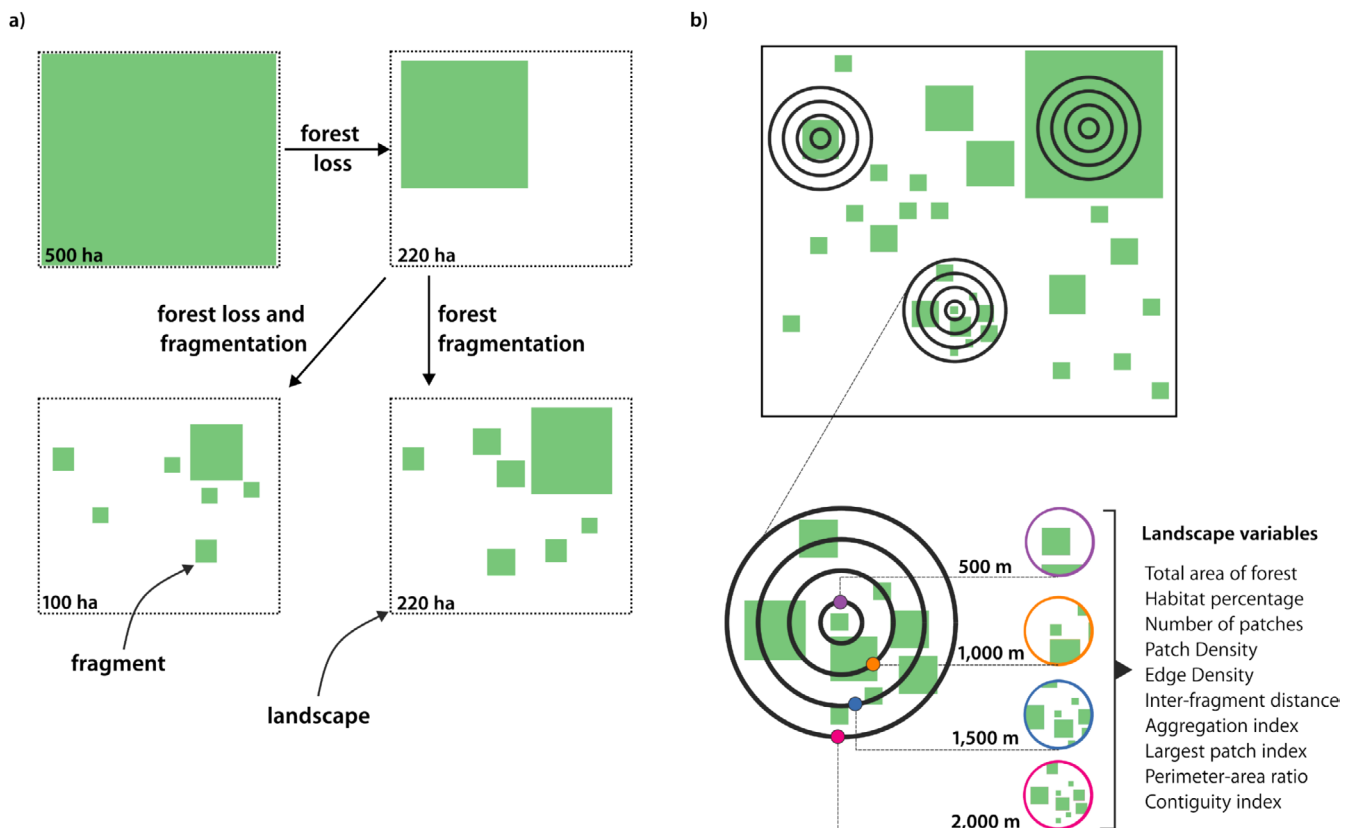


FIGURE 3 | (A) Schematic representation of forest loss and fragmentation. The green squares illustrate either continuous or fragmented forest within a given landscape. The area, measured in hectares (ha), represents the total forest cover present in that landscape. (B) Illustrative description of the buffers used to evaluate the landscapes surrounding the focal fragments. Buffer size varies from 200 to 2000 m of radius (with 100 m intervals), and variables were estimated based on each corresponding buffer size (without scale). We illustrated only four buffers with a radius of 500, 1000, 1500 and 2000 m.

the current biodiversity crisis in human-dominated landscapes threatens the sustainability of natural ecosystems and our own survival (Cardinale et al. 2012). To predict, prevent and if possible, reverse such anthropogenic damage, ecological research has been evaluating the relative importance of different threats to biodiversity in these landscapes. Although the theoretical advances in the matter have been very fruitful (Ewers and Didham 2006; Tscharntke et al. 2012; Fahrig 2017), empirical evidence is still limited and often debated. The LandFrag dataset is a global and up-to-date resource that can be used to carry out comprehensive assessments of these and other interesting theoretical models, which are of great applied value. Other potential uses of this database include the assessment of (1) the taxonomic and functional structure of communities in human-modified landscapes, (2) the functional signal of the responses of species to forest disturbances across scales and forest types, (3) the relationship between taxonomic and functional diversity metrics and (4) the taxonomic and functional predictors of the scale of landscape effects.

This dataset has a wide range of applications in biodiversity research and can be used to address a multitude of intriguing ecological and conservation questions at all spatial scales, from local to global. For example, for decades, ecologists and conservation biologists have noted that species diversity is usually lower in smaller and edge-affected forest fragments (Haddad et al. 2015; Fletcher et al. 2018). However, recent evidence suggests that species density in a given site can be more strongly related to habitat amount (e.g., forest cover) in the surrounding landscape than to the size of the fragment where the site is located (Watling et al. 2020). The LandFrag dataset can be used to untangle in what situations fragment- and landscape-scale variables determine biodiversity change. For instance, one potential source of confusion when comparing studies performed in different regions is that the effect of forest loss can vary depending on the species (Davies et al. 2004; Newbold et al. 2013; Pfeifer et al. 2017; Watling et al. 2020; Saldivar-Burrola et al. 2022) and the spatial extent (landscape size) at which forest cover is measured (Jackson and Fahrig 2015). This is also one approach that could be tested with LandFrag. Because this new dataset provides traits for many species, it is also possible to investigate which traits influence whether a given species will persist in fragmented landscapes, a topic that remains poorly investigated (Davies et al. 2004; Pinho et al. 2024).

Lastly, there is a relevant debate in the literature about the relative effects of forest loss and fragmentation on species and trait diversity. In fact, the effect of forest fragmentation on biodiversity remains contentious, since positive, negative and weak effects have been reported in the literature, and in many cases, fragmentation effects are confounded with the effect of forest loss (Chase et al. 2020; Riva et al. 2024; Zhang et al. 2024). Therefore, the LandFrag dataset should stimulate future research on forest loss and fragmentation, as well as their effects on biodiversity, providing new directions and potential solutions for biodiversity conservation (Gonçalves-Souza et al. 2025).

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The dataset and code can be downloaded on Zenodo <https://zenodo.org/records/12206838> or GitHub <https://github.com/mauriciovancine/landfrag>.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.